Assignment 1 – Probability and Statistics

Course Code: BBS00014

Question 1. *Find the correlation coefficient from the following data:*

X	6	2	10	4	8
Y	9	11	5	8	7

Solution:

Question 1. Find the correlation coefficient from the following data.

Formula :

$$\boxed{r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{\left[n\sum X^2 - (\sum X)^2\right]\left[n\sum Y^2 - (\sum Y)^2\right]}}}$$

Given data :

	X	Y	X^2	Y^2	XY
	6	9	36	81	54
	2	11	4	121	22
	10	5	100	25	50
	4	8	16	64	32
	8	7	64	49	56
Σ	30	40	220	340	214

Therefore,

$n = 5$ $\quad$ $\sum X = 30$ $\quad$ $\sum Y = 40$

$\sum X^2 = 220$ $\quad$ $\sum Y^2 = 340$ $\quad$ $\sum XY = 214$

Now,

$$r = \frac{5(214) - (30)(40)}{\sqrt{\left[5(220) - (30)^2\right]\left[5(340) - (40)^2\right]}}$$

$$= \frac{1070 - 1200}{\sqrt{(1100 - 900)(1700 - 1600)}}$$

$$= \frac{-130}{\sqrt{(200)(100)}}$$

$$= \frac{-130}{\sqrt{20000}}$$

$$= \frac{-130}{141.421}$$

$$= -0.919$$

Thus, $\boxed{r \approx -0.92}$

Hence, the correlation coefficient between X and Y is -0.92.

This shows a strong negative correlation between X and Y.

Question 2. *From a standard deck of 52 playing cards, two cards are drawn simultaneously. Calculate the probability that exactly one of the two cards is a face card (Jack, Queen, or King).*

Solution:

Question 2. From a standard deck of 52 playing cards, two cards are drawn simultaneously. Calculate the probability that exactly one of the two cards is a face card (Jack, Queen, or King).

Given :

⇒ Total number of cards in a deck, $n = 52$

⇒ Face cards (J, Q, K) in a deck $= 3 \times 4 = 12$

⇒ Non-face cards in a deck $= 52 - 12 = 40$

⇒ Two cards are drawn simultaneously.

Category	Number of cards
Face cards (J, Q, K)	12
Non-face cards	40
Total cards	52

Required : Probability that exactly one card is a face card.

Solution : Exactly one face card means one face card and one non-face card.

Number of favourable ways :

Choose 1 face card from 12 face cards and 1 non-face card from 40 non-face cards.

$$n(E) = {}^{12}C_1 \times {}^{40}C_1 = 12 \times 40 = 480$$

Total number of possible ways (choosing any 2 cards from 52) :

$$n(S) = {}^{52}C_2 = \frac{52!}{2!\,(52-2)!} = \frac{52 \times 51}{2 \times 1} = 1326$$

Therefore, the required probability,

$$P(\text{exactly one face card}) = \frac{n(E)}{n(S)} = \frac{480}{1326} = \frac{80}{221} \approx 0.362$$

$$\boxed{\therefore\ P(\text{exactly one face card}) = \frac{80}{221} \approx 0.362 \text{ (or } 36.2\%)}$$